

AUTOMATED INCIDENT RESPONSE SYSTEM



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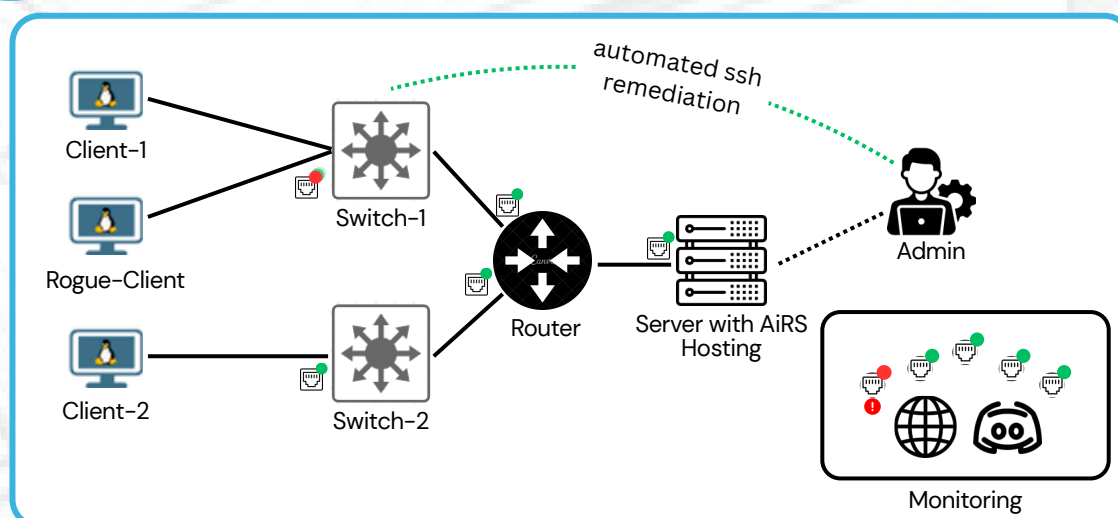
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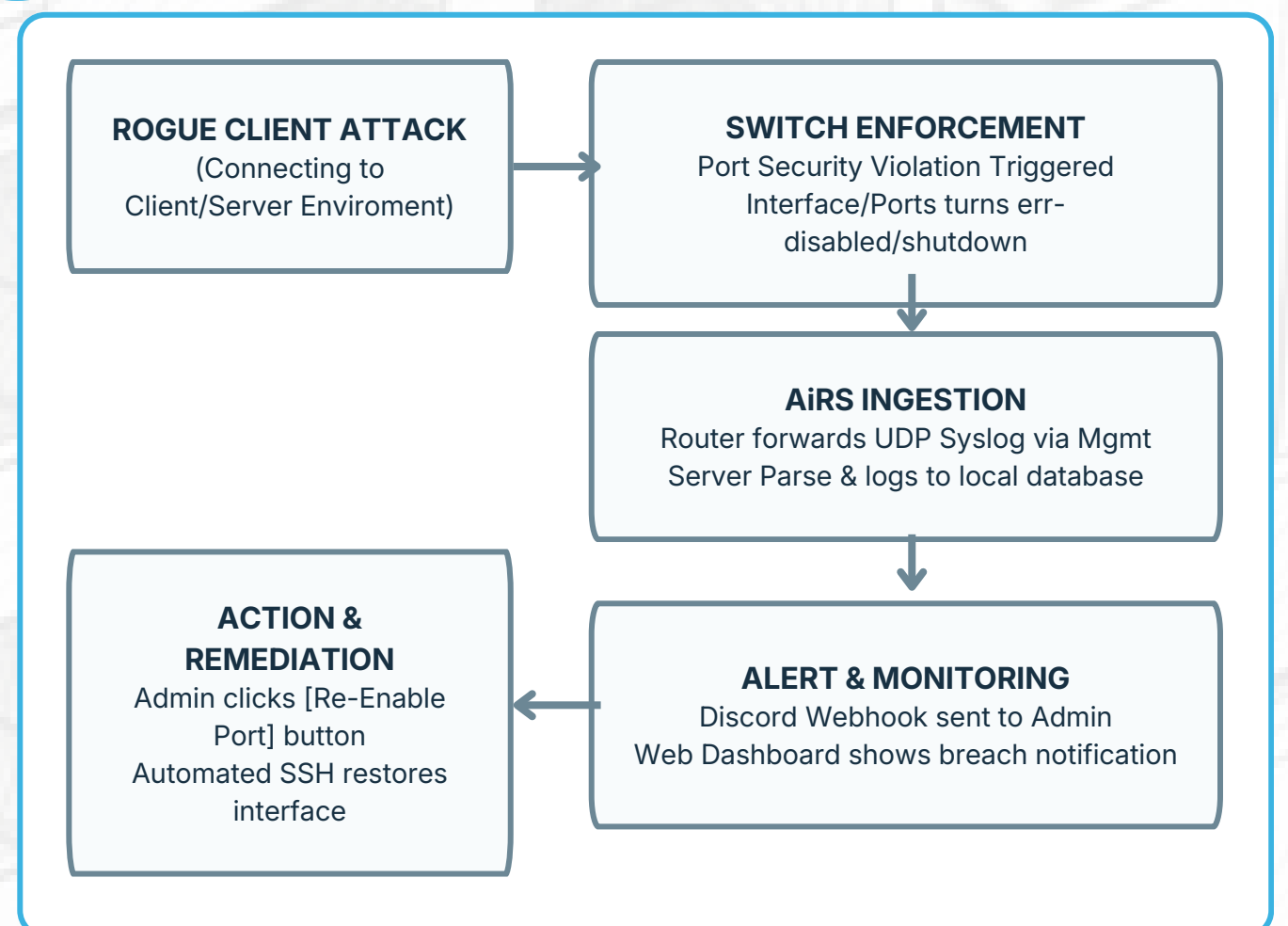
PROJECT OVERVIEW

Enterprise network security relies on Cisco Port Security to block unauthorized devices by locking ports into an err-disabled state upon detecting a violation. However, traditional enterprise automation tools often **require expensive software licensing**, while **manual recovery** forces administrators to rely on CLI-based SSH commands, leading to **extended network downtime**. The Automated Incident Response System (AiRS) solves this by providing a cost-effective solution that **captures real-time UDP Syslog alerts** and **features a centralized web dashboard** for instant, **one-click automated port remediation**.

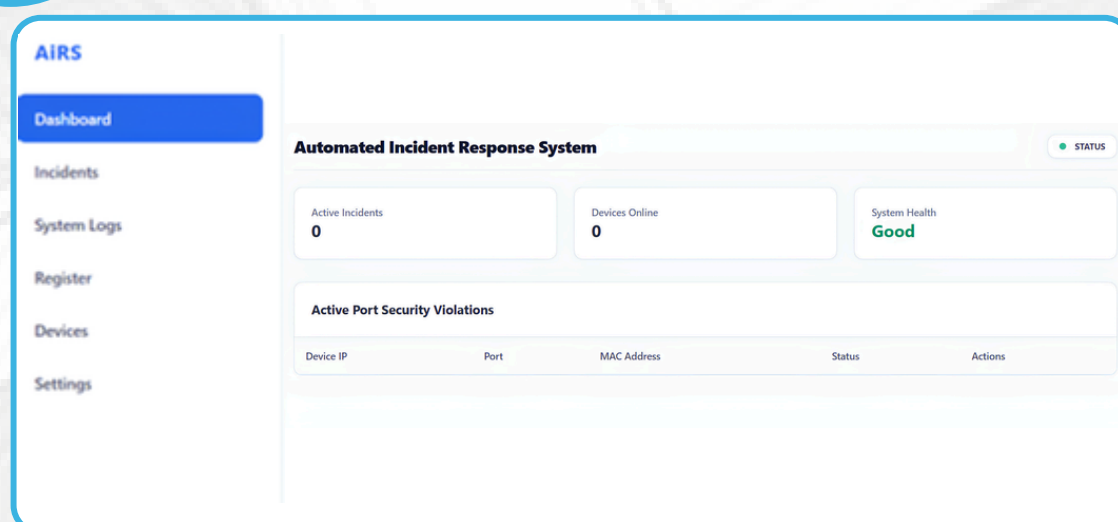
1 TOPOLOGY



2 SYSTEM FLOW



3 WEB DASHBOARD



4 SOFTWARE & HARDWARE



- Cisco (Router & Swtiches)
- EVE-NG (Emulation)
- VMware (Virtualization)
- VSCode
- Gemini

5 FUTURE IMPROVEMENT

Wireless Security Expansion (Cisco ISE)

Cisco ISE can be deployed as a virtual node alongside switches and routers in EVE-NG. This extends automated security and MAC enforcement to wireless networks via 802.1X authentication, centralized RADIUS policies, and dynamic threat quarantining.

Backend System Isolation (VMware Environment)

Hosting the entire AiRS backend including engine.py, app.py, and the database within an isolated VMware Windows Server VM optimizes system performance. Dedicated virtualization allows precise allocation of CPU cores and RAM, creating a cleaner and resource-optimized.

